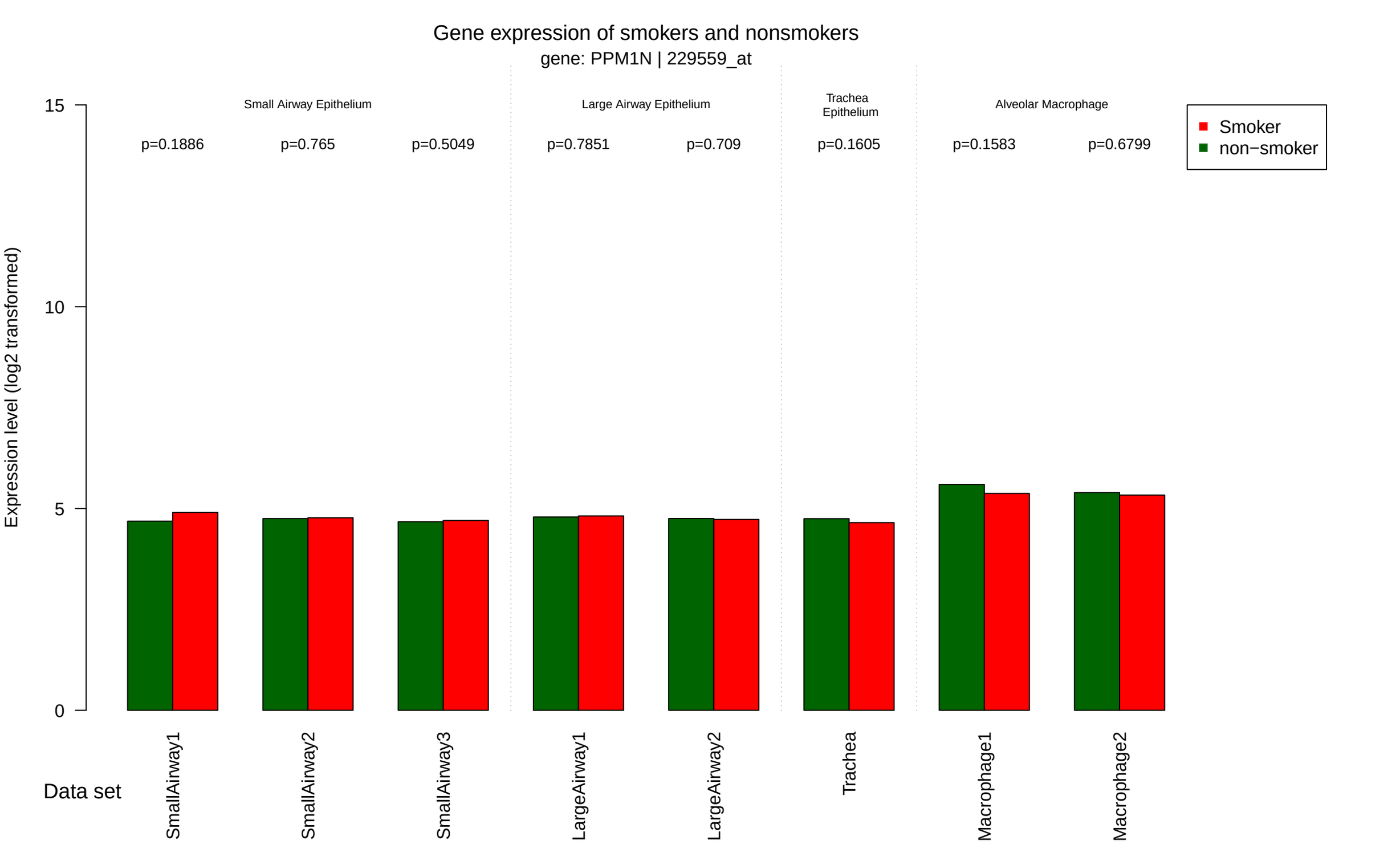
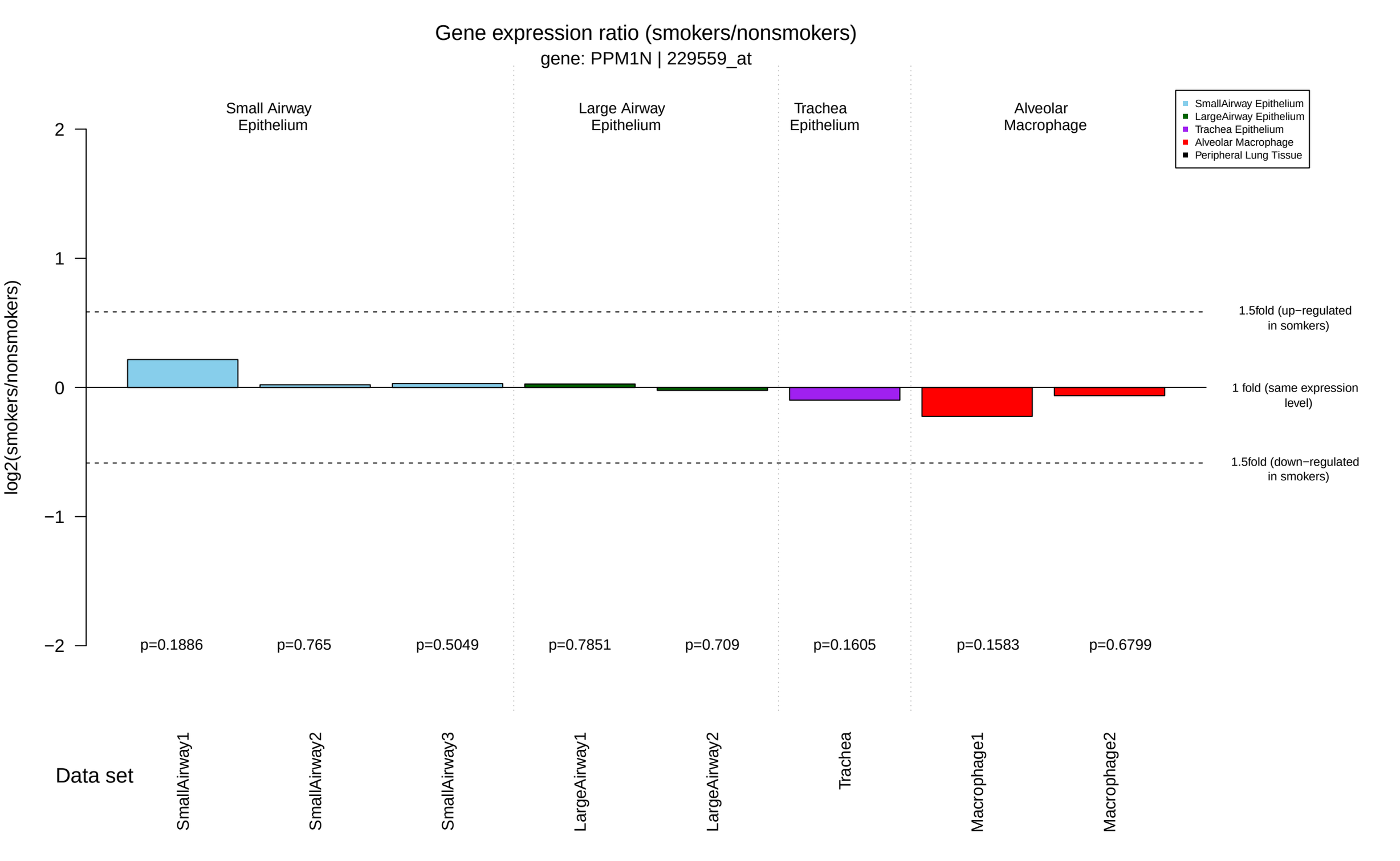
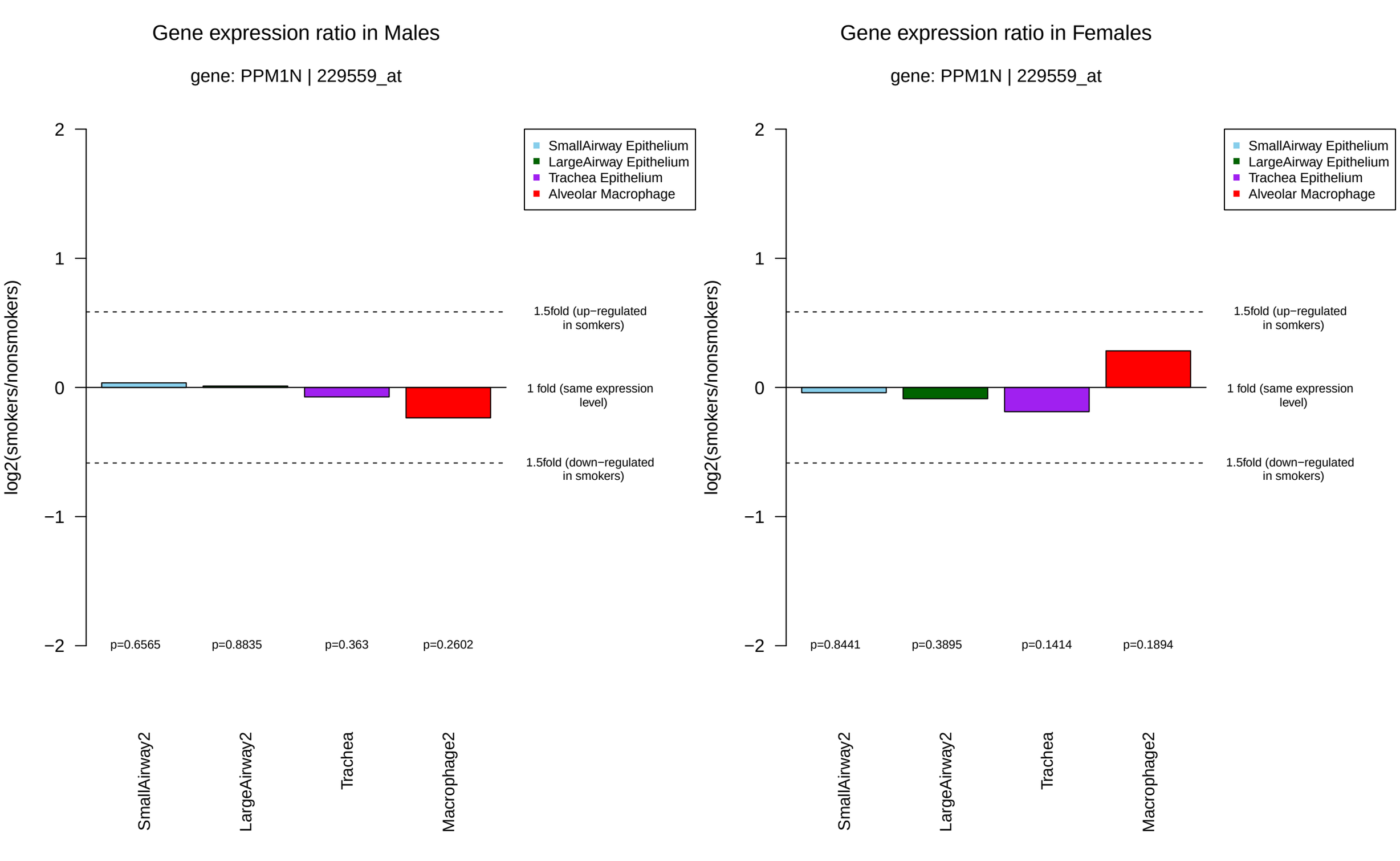
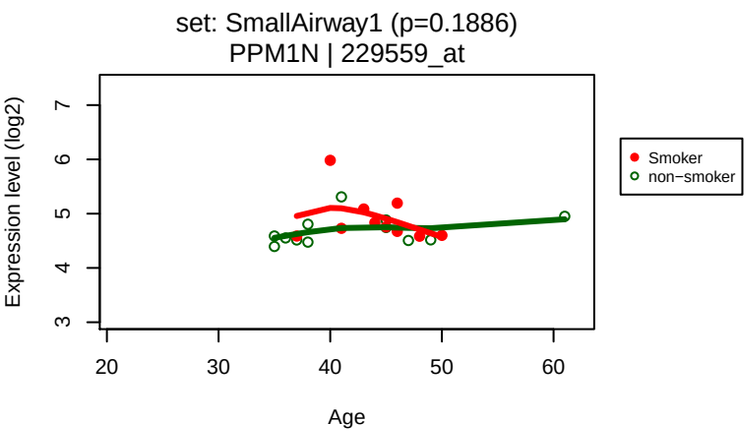
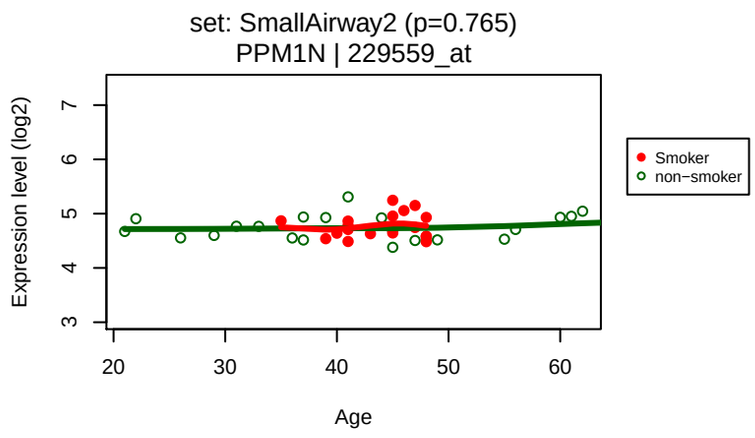
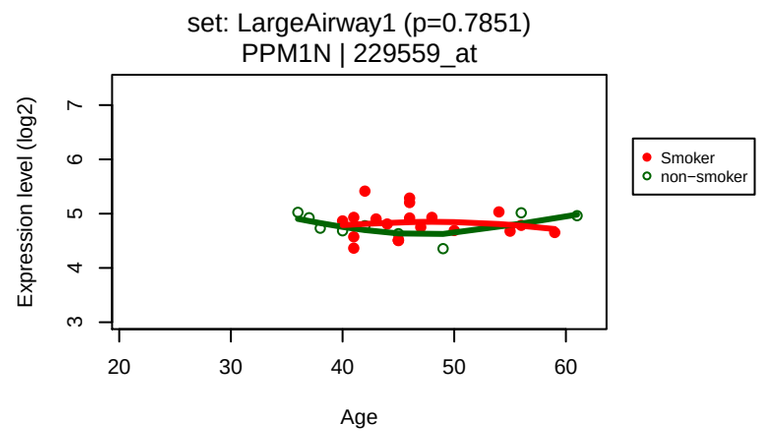

Gene expression of smokers and nonsmokers
gene: PPM1N | 229559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1886
p=0.765
p=0.5049
p=0.7851
p=0.709
p=0.1605
p=0.1583
p=0.6799
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

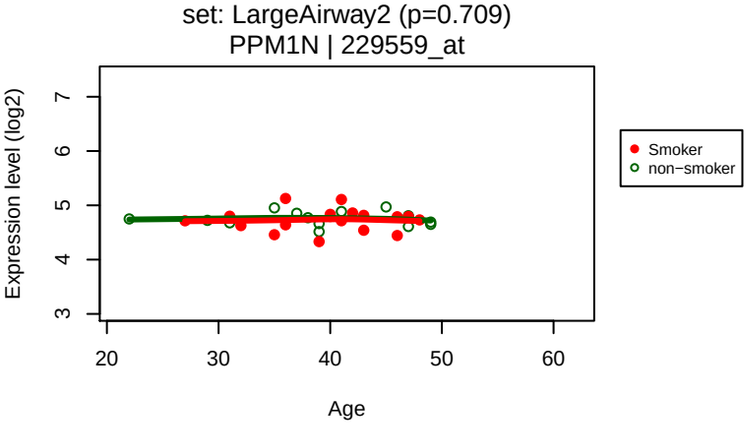

Gene expression ratio (smokers/nonsmokers)
gene: PPM1N | 229559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1886
p=0.765
p=0.5049
p=0.7851
p=0.709
p=0.1605
p=0.1583
p=0.6799
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

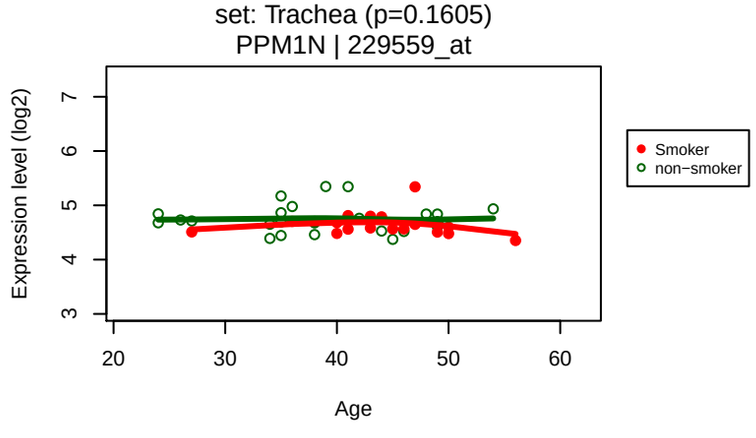

Gene expression ratio in Males
gene: PPM1N | 229559_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6565
p=0.8835
p=0.363
p=0.2602
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1N | 229559_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8441
p=0.3895
p=0.1414
p=0.1894
SmallAirway2
LargeAirway2
Trachea
Macrophage2

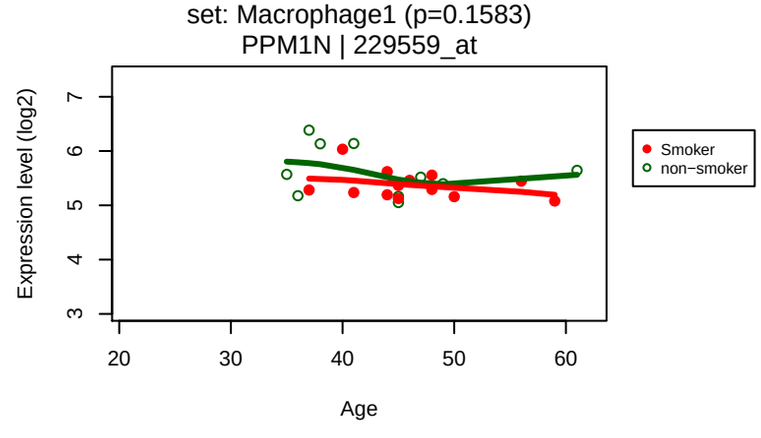

set: SmallAirway1 (p=0.1886)
PPM1N | 229559_at
Expression level (log2)
Age
Smoker
non−smoker

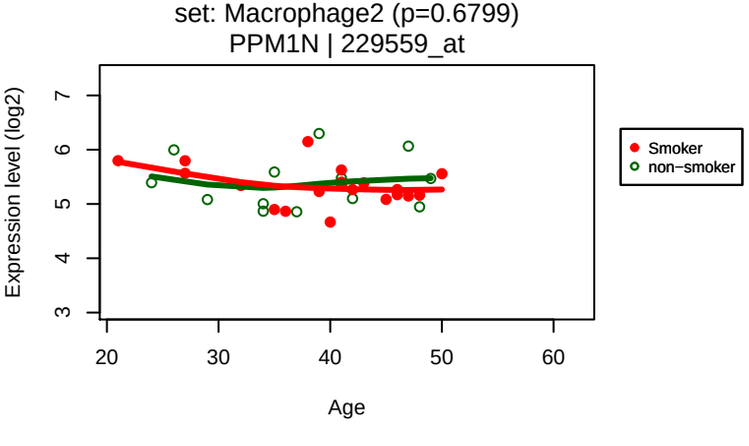

set: SmallAirway2 (p=0.765)
PPM1N | 229559_at
Expression level (log2)
Age
Smoker
non−smoker

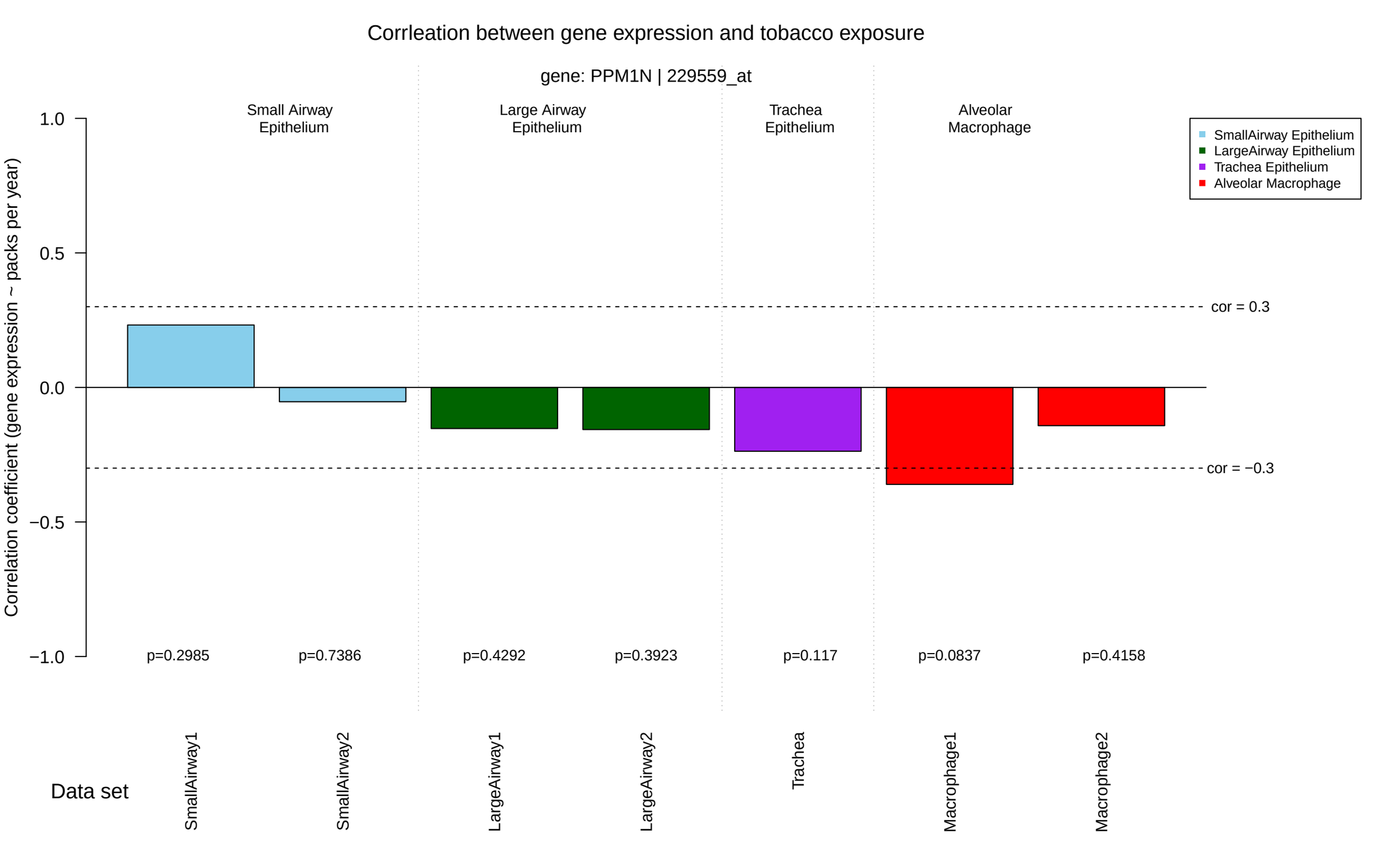

set: LargeAirway1 (p=0.7851)
PPM1N | 229559_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.709)
PPM1N | 229559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1605)
PPM1N | 229559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1583)
PPM1N | 229559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6799)
PPM1N | 229559_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1N | 229559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2985
p=0.7386
p=0.4292
p=0.3923
p=0.117
p=0.0837
p=0.4158
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2